

MARSHALL, AK, USA

WMO: 702628

Lat: 61.867N

Lon: 162.033W

Elev: 102

StdP: 14.64

Time Zone: -9.00 (NAK)

Period: 06-19

WBAN: 00105

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.3	-20.5	-31.3	0.9	-23.7	-27.0	1.2	-19.8	43.2	20.9	40.7	18.8	3.2	70	0.837

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.4	74.7	59.4	70.8	58.0	66.3	56.3	61.5	71.2	59.3	68.1	57.4	64.4	9.4	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.2	70.1	64.1	55.3	65.4	61.8	53.9	62.1	60.5	27.4	71.1	25.9	68.4	24.7	64.4	72.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	35.0	31.0	26.7	DB	-34.8	81.3	7.2	5.4	-40.0	85.2	-44.2	88.3	-48.2	91.4	-53.4	95.3
(5)				WB	-30.5	62.9	4.3	3.0	-33.7	65.0	-36.2	66.7	-38.6	68.4	-41.8	70.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.5	10.1	13.7	15.5	29.4	42.6	52.1	55.0	52.7	45.5	33.0	16.0	11.3
	DBStd	20.46	18.25	15.61	13.59	11.48	8.18	5.67	5.21	5.11	6.51	8.80	13.61	15.52
	HDD50	7175	1236	1016	1071	618	251	38	12	28	157	527	1021	1200
	HDD65	12232	1701	1436	1536	1067	696	390	314	381	584	991	1471	1665
	CDD50	425	0	0	0	0	20	101	167	113	23	1	0	0
	CDD65	7	0	0	0	0	0	2	4	1	0	0	0	0
	CDH74	145	0	0	0	0	3	45	88	9	0	0	0	0
	CDH80	24	0	0	0	0	0	9	15	0	0	0	0	0
Wind	WSAvg	11.3	16.1	15.3	11.8	10.1	9.3	8.2	8.6	8.5	9.5	11.3	11.8	14.7
Precipitation	PrecAvg	24.3	1.7	1.5	1.3	1.3	1.8	1.8	2.7	3.1	3.0	2.4	2.4	1.9
	PrecMax	30.2	3.7	4.4	4.9	2.7	2.9	3.1	4.7	5.6	5.6	5.1	5.4	4.3
	PrecMin	18.3	0.5	0.3	0.3	0.4	0.4	0.6	1.0	1.1	0.7	0.4	0.1	0.8
	PrecStd	3.3	0.8	1.0	0.9	0.6	0.6	0.5	0.8	0.9	1.1	1.1	1.3	0.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	43.8	40.7	44.5	56.9	73.0	78.9	82.0	74.9	64.3	53.6	42.6	45.4
		MCWB	38.9	36.8	38.2	44.3	55.4	60.2	64.1	60.8	53.5	46.9	39.4	37.4
	2%	DB	39.3	37.6	38.8	52.6	65.4	73.6	75.9	69.8	60.8	50.0	39.7	38.8
		MCWB	35.7	34.1	34.3	42.2	51.3	57.6	61.3	58.6	52.0	45.6	37.6	35.2
	5%	DB	36.9	35.9	36.1	48.4	61.1	69.7	71.6	65.7	57.1	47.6	37.1	35.4
		MCWB	34.4	33.1	32.9	40.4	49.1	56.2	60.1	56.6	51.1	44.7	35.6	33.3
	10%	DB	33.8	33.8	33.7	45.1	56.8	64.9	66.3	63.1	54.9	45.2	34.2	32.5
		MCWB	31.9	31.9	31.1	38.4	47.0	54.1	57.8	55.2	50.5	42.7	32.9	31.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.0	37.2	38.7	45.4	56.6	61.4	67.3	62.4	55.6	48.5	40.3	41.1
		MCDB	43.1	40.3	43.9	55.4	70.1	75.0	79.1	71.1	59.2	51.5	41.9	43.9
	2%	WB	36.3	34.7	35.0	42.6	52.8	59.3	63.1	60.0	53.6	46.4	37.5	35.4
		MCDB	39.0	36.9	37.8	51.4	64.8	71.3	73.9	67.1	58.8	48.9	39.0	37.9
	5%	WB	34.4	33.3	33.1	40.7	49.8	56.8	60.4	58.1	52.0	44.9	35.6	33.4
		MCDB	36.8	35.2	35.6	48.2	59.8	66.9	69.4	64.0	56.3	47.2	37.2	35.4
	10%	WB	32.1	31.9	31.0	38.6	47.7	54.8	58.3	56.3	50.7	42.8	32.9	31.0
		MCDB	33.8	33.7	33.3	44.6	55.9	64.4	65.8	61.4	54.6	45.4	34.1	32.7
Mean Daily Temperature Range	5% DB	MDBR	11.1	13.4	15.5	16.4	18.4	18.4	15.4	14.6	13.4	10.5	11.0	11.6
		MCDBR	9.4	10.0	13.5	20.0	26.7	28.7	27.7	22.5	17.1	10.6	8.0	10.5
		MCWBR	8.3	8.5	10.5	12.4	14.8	14.7	15.4	12.6	10.7	8.0	7.4	9.1
	5% WB	MCDBR	9.9	10.4	12.0	19.1	24.5	24.8	24.0	18.0	14.8	9.5	7.9	9.5
		MCWBR	8.8	9.3	9.7	12.1	13.9	13.3	13.9	10.7	9.9	7.7	7.4	8.5
Clear-Sky Solar Irradiance	taub	0.217	0.275	0.280	0.351	0.373	0.365	0.367	0.384	0.331	0.291	0.230	0.200	
	taud	2.038	2.156	2.230	2.087	2.279	2.386	2.421	2.353	2.486	2.400	2.042	1.887	
	Ebn at Noon	153	217	266	262	268	273	269	252	249	216	140	89	
	Edh at Noon	11	21	29	41	37	34	32	31	23	17	11	7	
All-Sky Solar Radiation	RadAvg	100	322	909	1434	1637	1686	1378	1087	730	354	118	47	
	RadStd	14	41	88	117	156	167	119	129	71	29	19	5	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (11 neighbors)	N/A	+3.01	+2.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page